New York State Finances: 1813 to 1860*

GJH† TJS‡

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1 Summary

- This document reports tables and figures that might find their way into a chapter on early state finance. New York might be the "good" case study. Pennsylvania might be the "bad" case study.
- In the 1790s and early 1800s, New York legislators aspired to finance state government expenditures through permanent funds derived from land sales and special duties. That is, they wished to create a commonwealth without taxation. Later, with the opening of the Erie Canal, New York became largely a venture capital operation.
- New York paid off its debt in the 1820s but had resume borrowing to pay for ordinary expenditures in the 1830s.
- New York borrowed heavily to finance the building and maintenance of the Canal System.
- In the late 1830s and early 1840s, New York State faced two major fiscal shocks.
    1. Three railroad companies whose debt was guaranteed by the State went bankrupt.
    2. A state-run bank insurance system collapsed when 11 banks failed.
- Despite these two shocks, New York State did not default on its debt during the State Debt Crisis of the 1840s. Nevertheless, its bonds traded at a discount during this period.
- Like many other states, New York adopted in a new constitution after the crisis of the early 1840s.
- There are connections to Hamilton's 1790 refunding. There are unit of account issues.

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†Brandeis University, E-mail: ghall@brandeis.edu.

‡New York University, E-mail: thomas.sargent@nyu.edu.

2 Three Government Budget Constraints

The organizing principle is the government budget constraint with private assets (which we now label "state funds").

$$\underbrace{G_t}_{\substack{\text{government}\\\text{spending}}} + \underbrace{r_t^B B_{t-1}}_{\substack{\text{interest on}\\\text{state debt}}} + \underbrace{A_t - A_{t-1}}_{\substack{\text{change in}\\\text{state funds}}} = \underbrace{T_t}_{\text{taxes}} + \underbrace{r_t^A A_{t-1}}_{\substack{\text{interest on}\\\text{state funds}}} + \underbrace{B_t - B_{t-1}}_{\substack{\text{change in}\\\text{state debt}}} \tag{1}$$

- Equation (1) consolidates both the ordinary part of the government, which financed usual government expenditures (i.e., schools, courts), and the Canal Commission, which financed the Erie and other canals. We label these two parts the "General Account" and the "Canal Account."

- The budget constraint for the Canal Account is:

$$\begin{aligned}\underbrace{F_t^c - F_{t-1}^c}_{\substack{\text{change in}\\\text{Canal Fund}}} = {} & \underbrace{B_t^c - B_{t-1}^c}_{\text{new Canal Debt}} + \underbrace{TOLL_t^c}_{\text{tolls}} + \underbrace{T_t^c}_{\text{taxes}} + \underbrace{OR_t^c}_{\text{other revenue}} + \underbrace{TR_t^{gc}}_{\substack{\text{transfers from General}\\\text{to Canal Account}}} \\ & + \underbrace{r_t^A F_{t-1}^c}_{\substack{\text{interest earned}\\\text{on Canal fund}}} - \underbrace{EXP_t^c}_{\text{Canal expenses}} - \underbrace{r_t^B B_{t-1}^c}_{\substack{\text{interest on}\\\text{Canal debt}}} - \underbrace{TR_t^{cg}}_{\substack{\text{transfers from Canal}\\\text{to General Account}}}\end{aligned} \tag{2}$$

- The budget constraint for the General Account is:

$$\begin{aligned}\underbrace{F_t^g - F_{t-1}^g}_{\substack{\text{change in}\\\text{General Fund}}} = {} & \underbrace{B_t^g - B_{t-1}^g}_{\substack{\text{change in}\\\text{General Debt}}} + \underbrace{T_t^g}_{\text{taxes}} - \underbrace{TR_t^{gc}}_{\substack{\text{transfers from General}\\\text{to Canal Account}}} + \underbrace{r_t^A F_{t-1}^g}_{\substack{\text{interest earned}\\\text{on General Fund}}} \\ & - \underbrace{S_t^g}_{\substack{\text{General Account}\\\text{spending}}} - \underbrace{r_t^B B_{t-1}^g}_{\substack{\text{interest on}\\\text{General Fund debt}}} + \underbrace{TR_t^{cg}}_{\substack{\text{transfers from Canal}\\\text{to General Account}}}\end{aligned} \tag{3}$$

- Adding equations (2) and (3) yields:

$$\begin{aligned}\underbrace{(F_t^c + F_t^g) - (F_{t-1}^c + F_{t-1}^g)}_{\substack{\text{change in}\\\text{State Funds}}} = {} & \underbrace{(B_t^c + B_t^g) - (B_{t-1}^c + B_{t-1}^g)}_{\substack{\text{change in}\\\text{State Debt}}} + \underbrace{TOLLS_t^c + T_t^c + T_t^g + OR_t^c}_{\text{taxes}} \\ & + \underbrace{r_t^A(F_{t-1}^c + F_{t-1}^g)}_{\substack{\text{interest earned}\\\text{on State Funds}}} - \underbrace{(S_t^g + EXP_t^c)}_{\substack{\text{government}\\\text{spending}}} - \underbrace{r_t^B(B_{t-1}^c + B_{t-1}^g)}_{\substack{\text{interest on}\\\text{State debt}}}\end{aligned} \tag{4}$$

Noting that:

$$
\begin{aligned}
A_t &= F_t^c + F_t^g \\
B_t &= B_t^c + B_t^g \\
T_t &= TOLLS_t^c + T_t^c + T_t^g + OR_t^c \\
G_t &= S_t^g + EXP_t^c
\end{aligned}
$$

we see that equation (4) restates equation (1).

3 New York State Funds

In the 1780s, New York State had the following debts

Description	Principal (£ s d)	Interest (£ s d)
Certificates for moneys loaned pursuant to Act of 1788	111 13 3	78 14 5
Certificates for moneys loaned pursuant to Act of 1780	741 6 0	472 10 9
For horses bought, 1780	904 5 0	515 8 5
For depreciation of pay to August 1780	54,520 1 7	25,669 17 4
Army pay for 1781	17,972 6 9	8,626 14 0
Pensions to officers	8,104 8 2	3,647 4 2
Pay of levies, militia	42,871 4 3	18,220 5 3
Received on loan from U.S., April 18, 1786	523,848 5 1	144,058 5 4
For 4/5 of interest due in this loan	105,669 9 8	—
Claims on forfeited estates	25,897 8 10	3,884 12 3
Bills of credit issued June 30, 1780	3,612 16 0	1,174 3 1
Miscellaneous	1,047 0 0	—
Total	785,300 14 7	206,297 15 0
Demands against forfeited estates (no certificates issued)	41,017 12 5	
Interest on above	206,297 15 0	
Total Liabilities	1,032,616 2 0	

It also held the following Continental Securities.

State Holdings of Continental Securities	Amount (£ s d)
Loan Office Certificates	360,683 9 3
Certificates, Wm. Barber	458,213 4 0
Certificates, Pierce, Burrel, Deming	389,498 9 9
Indent of Interest	2,502 14 8
Total	1,210,897 16 8

In 1791, Alexander Hamilton exchanged these securities for

United States 6% Stock	$898,847.03
United States deferred 6% stock	520,688.66
United States 3% Stock	680,681.70
Total	$2,100,197.39

These tables are from Sowers (1914, pages 255-6). The accounts between the State and the Secretary of the Treasury of the United States were finally settled in 1800.

These three stocks formed the basis of what became known as the *General Fund*. The Comptroller was made custodian of this fund and was empowered to invest it. Over time, the fund was augmented from sales of public lands and other sources.

In the 1790s and early 1800s, New York legislators envisioned a fiscal system in which the state could sustain itself without resorting to direct taxation. That is, the revenue from the General Fund and other funds, as well as the sale of public land in the western part of the state, could cover ordinary government expenses.

After the War of 1812, the state was compelled to introduce recurring taxes, notably on property and auctions, marking the gradual abandonment of the early republican aspiration to build a government "without taxation."

Between 1813 and 1860, there were six major funds and eight minor sinking funds.

1. The *General Fund* was created from the 1790 federal settlement and was augmented with asset sales; it helped fund ordinary government, but its principal was gradually drawn down and ultimately replaced by debt.

2. The *Canal Fund* was financed by the revenue of the Erie Canal and other canals and used to pay canal expenses and debts.

3. The *Common School Fund* was an endowment fund for public education.

4. The *Literature Fund* was an endowment fund for secondary education and libraries.

5. The *Bank Fund* was a bank insurance fund. It went insolvent after the collapse of 11 banks between 1840 and 1842..

6. The *US Deposit Fund* was created after the federal government distributed surplus federal funds to the states in 1837.

The eight minor sinking funds were created after the assumption of stocks from three failed railroads. I need to track down some more details about the creation of these funds.

In figure 1, I plot the balances for these New York State funds. In the remainder of this section, we discuss each of these funds in more detail. These notes come from Roberts (1897) and various *Annual Reports of the Comptroller*.

3.1 The General Fund

- As can been seen in figure 1, in 1813, the General Fund balance was $4.4 million. The income from the fund, together with receipts from salt and auction duties, was thought sufficient to maintain the operations of the State government.

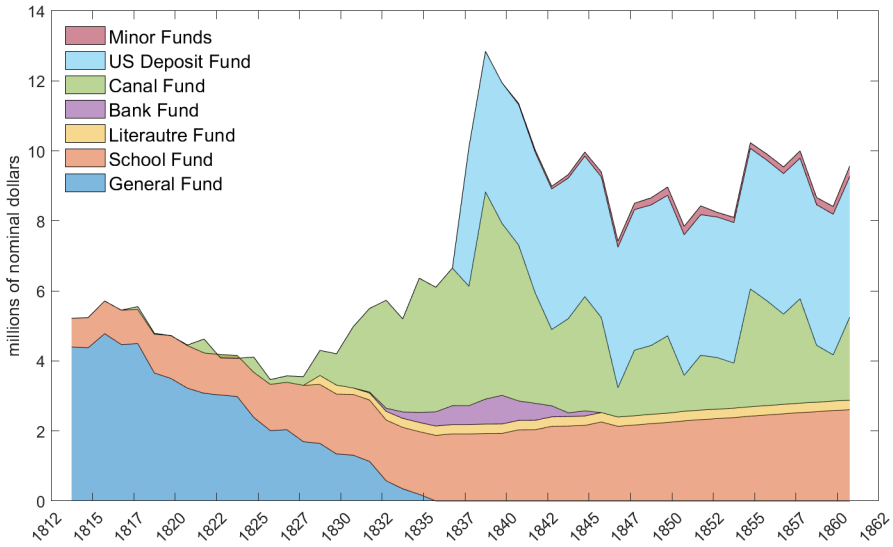

Figure 1: New York State Fund Balances

3.2 The Common School Fund

- In 1805, the Legislature used the proceeds from the sale of the first 500,000 acres of vacant and unappropriated lands managed by the Surveyor-General to establish the *Common School Fund*, a permanent fund for the support of common schools.

- This represented one of the earliest efforts by any American state to create an endowment explicitly dedicated to public education.

- Additional revenues were later added from various sources, and the fund grew from $0.82 million in 1813 to $2.6 million in 1860.

- No direct state tax for school purposes was levied until 1853. Thus, from 1805 to 1853, the interest of the Common School Fund, augmented by interest from the U.S. Deposit Fund (discussed below), was used to support the State's system of free public education. This policy reflected the early republican ideal that education, like government itself, might be sustained from the returns on public assets rather than from direct taxation.

3.3 The Literature Fund

- New York State Literature Fund was established in 1827 to support secondary education and the creation of libraries in academies, which were private schools for older students.

- Revenue for the Literature Fund came from various sources, including revenue from public lands and a small portion of the General fund.
- Unlike the Common School Fund, which supported local, free-tuition elementary schools, the Literature Fund primarily supported academies, which typically charged tuition and catered to a wealthier class.
- The Literature Fund was part of a larger state effort to provide public education. By funding libraries and teachers in academies, the fund promoted the study of literature and a broader curriculum beyond the basic literacy and moral instruction offered in the common schools.
- The Literature Fund was eventually consolidated with the United States Deposit Fund.

3.4 United States Deposit Fund

- The United States Deposit Fund was created after the federal government distributed surplus federal funds to the states in 1837.
- The money in the fund was lent out by the state and was intended to be secured by real estate mortgages.
- The interest generated from these loans was used to support the state's common school fund.

3.5 The Bank Safety Fund

- In 1829, New York State set up a bank insurance system called the *Bank Safety Fund.*
- The system required banks to deposit securities in the Comptroller's office as a guaranty for the notes issued by the banks. It imposed annual assessments of 1/2 of 1% of a bank's paid-in capital until each bank had paid in a total of 3%. Banks were permitted to deposit up to 1/2 of their security in the form of bonds and mortgages.
- This fund would then be used to reimburse bank creditors in full, should a bank fail. These creditors were largely note holders. Notes in circulation were twice the size of deposits at a typical bank.
- Albert Gallatin noted:

> ... with the laudable intent to protect the community against partial failures, a 'safety fund' has been established by law ... which is applicable to the payments of the notes of any that may fail. This must have the tendency to encourage excessive

issues of paper, which could not be sustained if resting only on the credit of the bank by which they are made.

Cite Bodenhorn (1996) for this quote.

- There was also a bank supervision aspect to the system. The legislation creating the system authorized the appointment of three Bank Commissioners charged with visiting and examining the banks and reporting to the Legislature. The law also required every banking association to file semi-annual reports of its transactions with the Comptroller.

- Between February 1840 and October 1842, 11 banks failed. These banks are listed in table 1. In 1839, the balance (I assume measured at its par value) in the safety fund was $0.818 million. Part of the banks' deposits in the Bank Safety Fund were bonds and mortgages, which fell in value during this period, the market value of the fund fell at exactly the time the Comptroller needed to sell assets to reimburse bank creditors.

- The total cost to fund for insuring the creditors for the 11 banks was $2.565 million. The fund became insolvent in 1843.

- The office of the Bank Commissioners was abolished in 1843, and the responsibility for bank supervision was transferred to the Comptroller. In 1851, a separate Banking Department was established.

Failed Bank	Date of Failure	Notes Redeemed	Other Debts Paid	Receipts from Asset Sales	Balance Paid by Fund
City Bank of Buffalo	2/1840	$317,107		$99,996	$217,111
Wayne County Bank	12/1840	113,131	$16,078		129,209
Commercial Bank of New York City	9/1841	139,837	146,129	7,188	278,778
Bank of Buffalo	11/1841	435,540	149,241		584,781
Commercial Bank of Buffalo	11/1841	186,861	424,515	5,000	606,376
Commercial Bank of Oswego	12/1841	163,162	78,352	2,392	239,122
Watervliet Bank	3/1842	134,107	77,484	13,258	198,333
Clinton County Bank	4/1842	71,896	156,257		228,153
Bank of Lyons	9/1842	52,898	40,053	3,761	88,990
Lafayette Bank, NYC	2/1842	38			38
Oswego Bank	10/1842				
Not specified		725		6,482	
Total	$1,615,302	$1,088,109	$138,077	$2,565,334	

Table 1: Payments Made by the Bank Safety Fund to Reimburse Creditors of the Failed Banks

Source: Chaddock (1910, page 332) and Bodenhorn (1996, page 25).

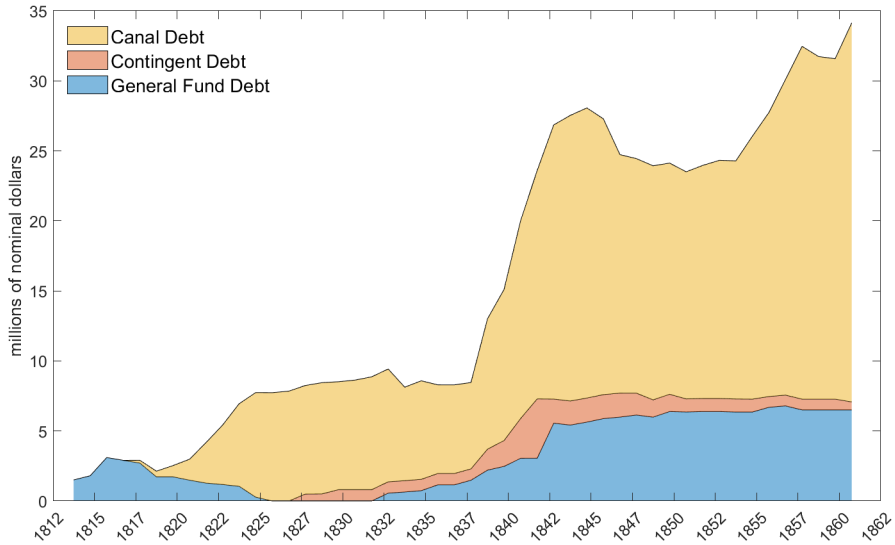

Figure 2: New York State Debt

3.6 Minor Funds

These minor funds were the Mariner's Fund, Auburn and Rochester Sinking Fund, Tonawanda Railroad Company Sinking Fund, Hudson & Berkshire Railroad Company Sinking Fund, Long Island Railroad Company Sinking Fund, Tioga Cola Iron Mining & Manufactures Sinking Fund, and the Stockbridge Indians Fund.

4 New York State Debts

In figure 2, I plot the quantity of New York State debt outstanding.

- The three types of debt: General Fund debt, Contingent debt, and Canal debt. Most of the debt issued by the New York was for the construction and maintenance of the Canal System.
- About 42% of the debt was held abroad. In table 2 we reproduce a table from the 1842 Annual Report of the Comptroller listing the bonds outstanding in 1841 and the distribution of their ownership.

4.1 General Fund Debt

- There is general fund debt in 1814 along with a positive balance in the General Fund. This debt was paid off by 1825.

	Held			
Bond	in New York	in Other States	by Foreigners	Total
Erie and Champlain Canals	$150,529.98	$18,495.00	$1,715,412.23	$1,884,437.21
Enlargement of Erie Canal	3,294,048.75	333,229.40	3,407,341.14	7,034,619.29
Cayuga, Seneca, and Oswego Canals	376,078.66	26,211.11	256,014.23	658,304.00
Crooked Lake Canal	114,000.00	—	5,500.00	120,000.00
Chemung Canal	156,006.93	5,700.00	314,707.30	476,974.23
Chenango Canal	629,232.31	223,392.14	1,549,911.21	2,402,535.66
Black River Canal	357,551.82	8,166.67	739,954.74	1,105,673.23
Genesee Valley Canal	606,063.82	81,905.00	1,878,411.00	2,566,379.82
Oneida River Improvement	43,800.00	1,200.00	5,000.00	50,000.00
Oneida Lake Canal	50,000.00	—	—	50,000.00
7 per cents of 1848	1,445,736.00	106,500.00	32,500.00	1,584,736.00
7 per cents of 1849	1,608,000.00	13,600.00	31,000.00	1,652,600.00
Delaware and Hudson Canal Co.	409,316.88	17,358.88	373,324.24	800,000.00
New York and Erie Railroad Co.	2,587,000.00	251,000.00	162,000.00	3,000,000.00
Tonawanda Railroad	86,000.00	14,000.00	—	100,000.00
Schenectady Railroad	100,000.00	—	—	100,000.00
Long Island Railroad	80,000.00	3,000.00	17,000.00	100,000.00
Catskill and Canajoharie Railroad	86,000.00	—	114,000.00	200,000.00
Ithaca and Owego Railroad	82,000.00	12,000.00	221,700.00	315,700.00
Auburn and Rochester Railroad	188,000.00	11,000.00	1,000.00	200,000.00
Hudson and Berkshire Railroad	141,000.00	—	9,000.00	150,000.00
Tioga Coal, Iron and Mining Co.	70,000.00	—	—	70,000.00
Astor Stock	561,500.00	—	—	561,500.00
5 per cent Bank Fund Stock	350,257.00	—	—	350,257.00
7 per cent stock and bonds	465,358.00	—	—	465,358.00
Total	14,038,540.15	1,240,962.10	10,833776.09	25,999,074.44

Table 2: Ownership Distribution of New York State Debts, 1841–42

Source: 1942 *Annual Report of the Comptroller*, pages 9-10.

- In 1832, the State resumed borrowing to pay for ordinary expenditures. The debt increases in the early 1840s as the government assumes the debts of three failed railroads. See the following subsection.

- The Constitution of 1846 introduced a new fiscal framework. It mandated the creation of a sinking fund for the gradual redemption of the General Fund debt. The management and investment of this sinking fund were entrusted to the Comptroller. The State retired the last of this debt in 1878.

4.2 Contingent Debt

- Starting 1827, the State issued debt on behalf of various canal and railroad companies. While the companies were responsible for making the coupon and principal payments, the authorizing legislation "pledges the faith and credit of this State for the payment of the interest and principal" on these bonds. These guarantees were backed by mortgages on the railroad and property.

- The coupon and principal payments were payable in London. (source is a Comptroller report in the early 1840s. Find it.)

- These debts were called *contingent debt*. They were implicit liabilities of the State.

- On July 1, 1841, the Ithaca and Owego, and Catskill and Canajoharie railroads stopped making their coupon payments. At a later date (find this date), the New York and Erie railroad defaulted as well. Consequently $3,315,700 in railroad debt became State obligations and part of the General Fund debt. This can be seen in figure 2 as between 1841 and 1842, the contingent debt drops by $2,515,700, and the General Fund debt increases by $2,503,324.14. There is an extra $1 million that must be "netted out" somehow. I need to figure this out.

4.3 Canal Debt

The Canal debt makes up the lion's share of the debt. It increased in early late 1810s and early 1820s as well as the early 1840s. These increase correspond to construction periods.

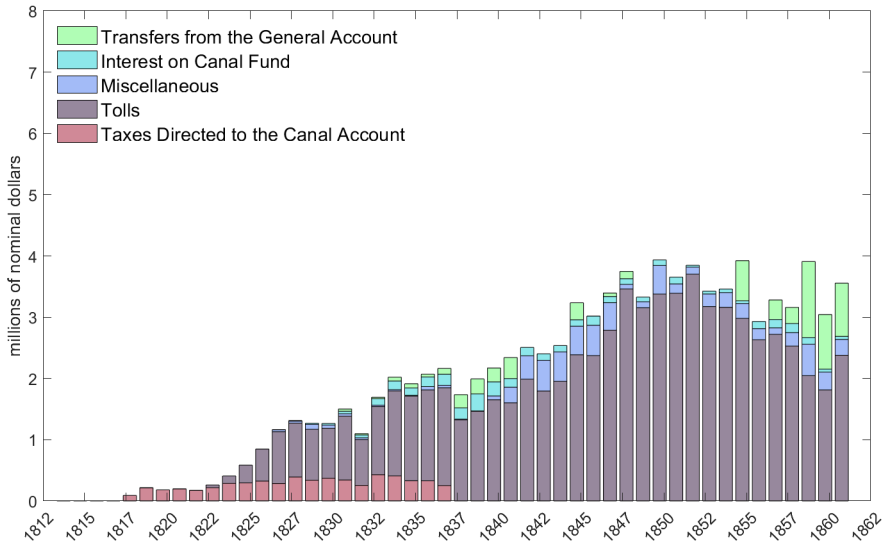

Figure 3: Canal Account Revenues

5 The Canal Account

In figures 3, 4, and 5, we plot the Canal Account's revenues and expenditures.

- In figure 3, we see that from 1817 to 1820, taxes were the sole source of revenue. These were taxes on auctions, salt, and steamboat revenue. Between 1817 and 1836, these taxes were diverted from the General Account to the Canal Account.
- While the Erie Canal officially opened on October 26, 1825, the Canal started to collect toll revenue in 1821 and by the late 1820s, it became the largest share of revenue. Revenues declined in the late 1830s and in the 1850s.
- In figure 4, we see that there are spikes in outlays due to the construction of new branches.
- In figure 5, we see a form of the "law of gravity" between outlays and revenues. I net out transfers to and from the General Fund in this figure.

In figure 6 I plot transfers between the Canal Fund and the General Fund. Prior to 1840, the Canal received transfers from the General Fund. After 1840, the flows went the other way.

The New York Canal System generated a tremendous amount of economic growth for the state. I don't have any data to support this, but it is clear that the New York State economy grew during this period. Ideally, I would plot these series as ratios of state GDP.

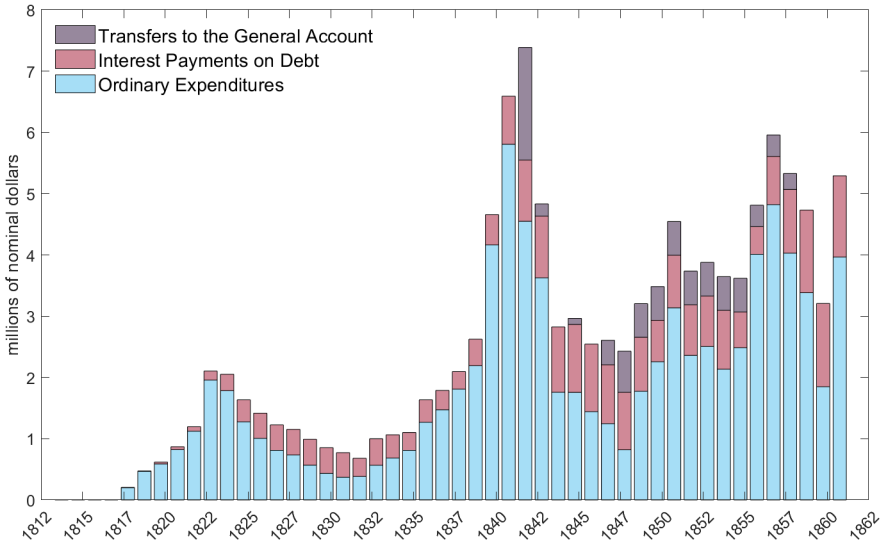

Figure 4: Canal Account Expenditures

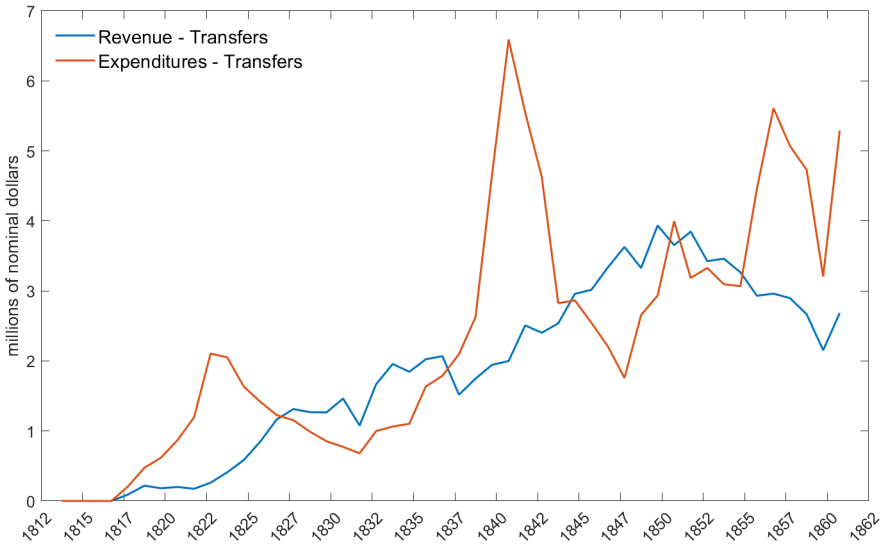

Figure 5: Canal Account Revenue and Expenditures

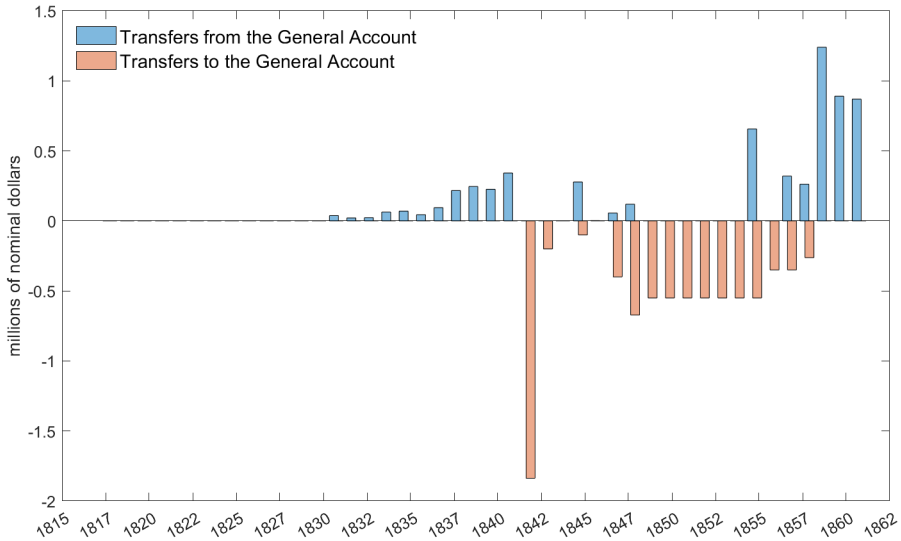

Figure 6: Transfers Between the Canal Accounts and the General Account

6 The General Account

- In figure 7, I plot the sources of General Account revenues from 1813 to 1860
  - Land sales make up a relatively small share of total revenue
  - During the late 1820s and early 1830s, revenue from the state accounts made up the lion's share of revenue. In 1832, the State resumed borrowing to pay for ordinary expenditures.
  - A state property tax was in effect between 1816 and 1826, which accounted for a large share of revenue. The property tax was reinstated in 1842.
  - The large payment from the federal government is the distribution of the federal surplus to the states mandated by the Deposit Act of 1836.

  In figure 8, I plot expenditures over time.

  - Ordinary expenditures peak during the War of 1812 and then decline. They generally rise over time. I don't scale by population or by output.
  - The assumption of the contingent debt from the three failed railroads (in purple) in 1841 and 1842 stands out.
  - The insurance payments to the creditors of the failed banks also raised expenditures in the early 1840s.

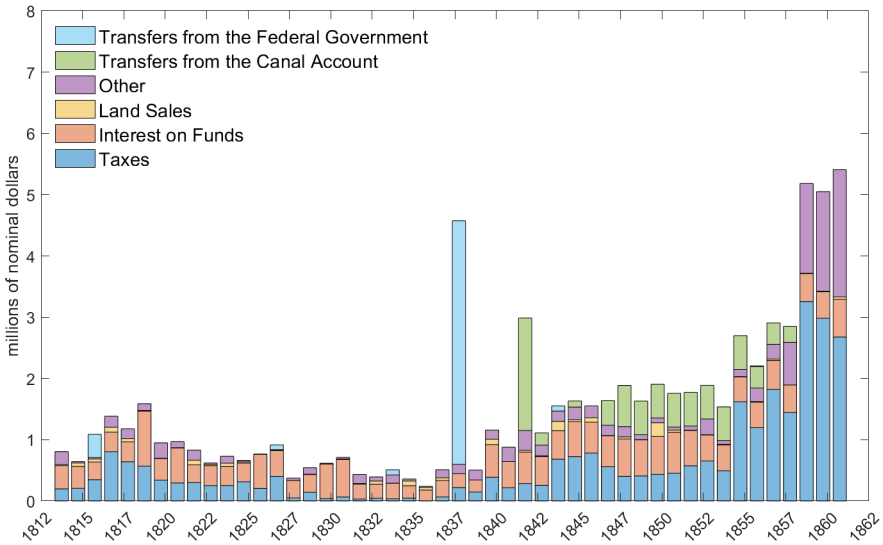

Figure 7: General Account Revenues

In figure 9 we plot General Account revenue and expenditures (net transfers) on a single figure. Before the early 1830s, the General Account's budget was balanced. In the 1840s and 1850s, the General Account ran deficits.

7 Consolidated Accounts

In figures 10, 11, and 12, we plot the merged Canal and General Account's revenues and expenditures.

- The Canal Account's revenues and expenditures are far larger than those of the General Account. New York State is primarily a canal operator with a secondary role in general government.
- With the big exception of the late 1830s and early 1840s, the consolidated budget is largely balanced over this time period.

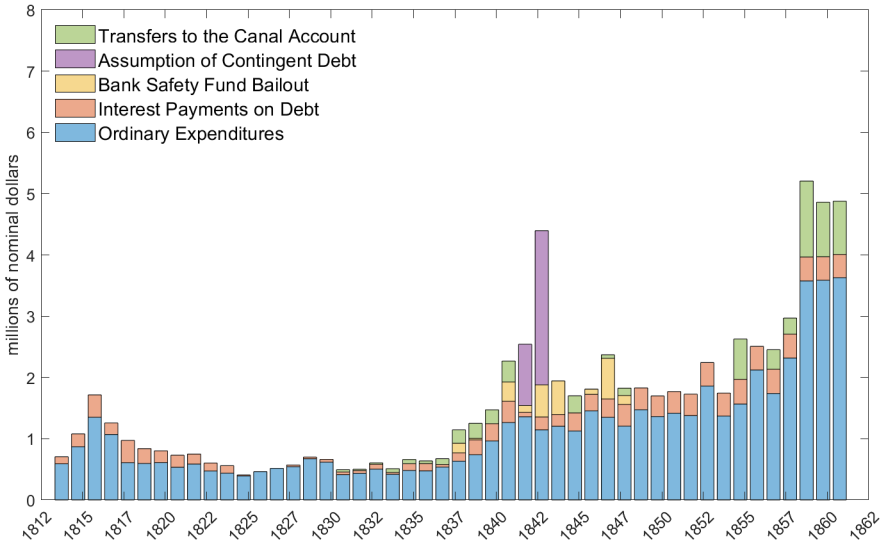

Figure 8: General Account Expenditures

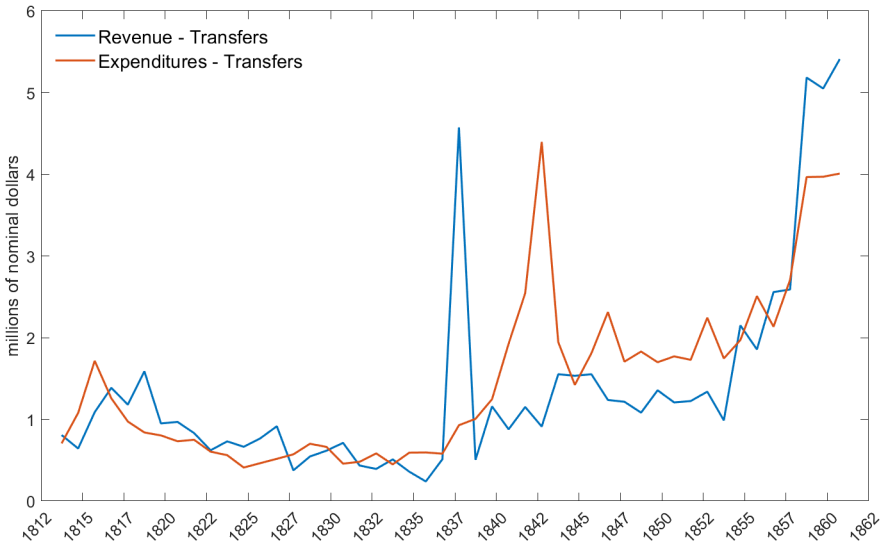

Figure 9: General Account Revenue and Expenditures

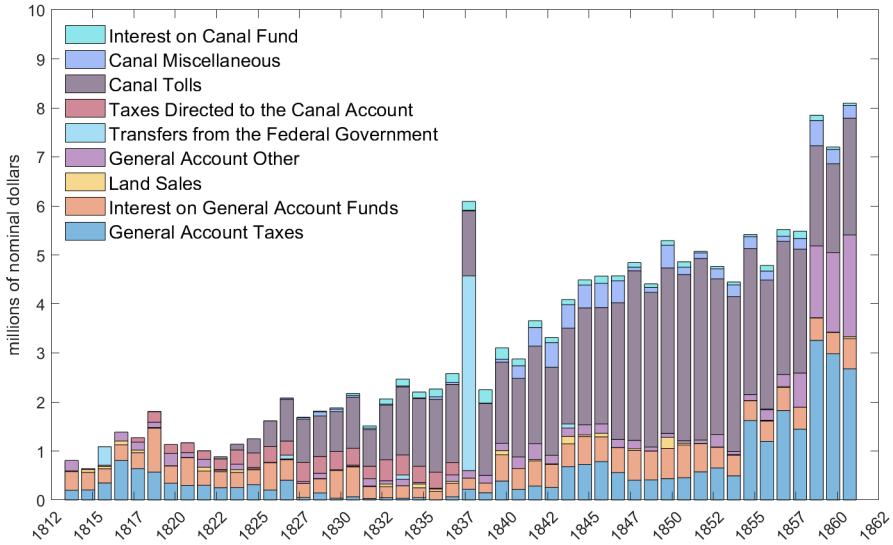

Figure 10: Consolidated Revenues

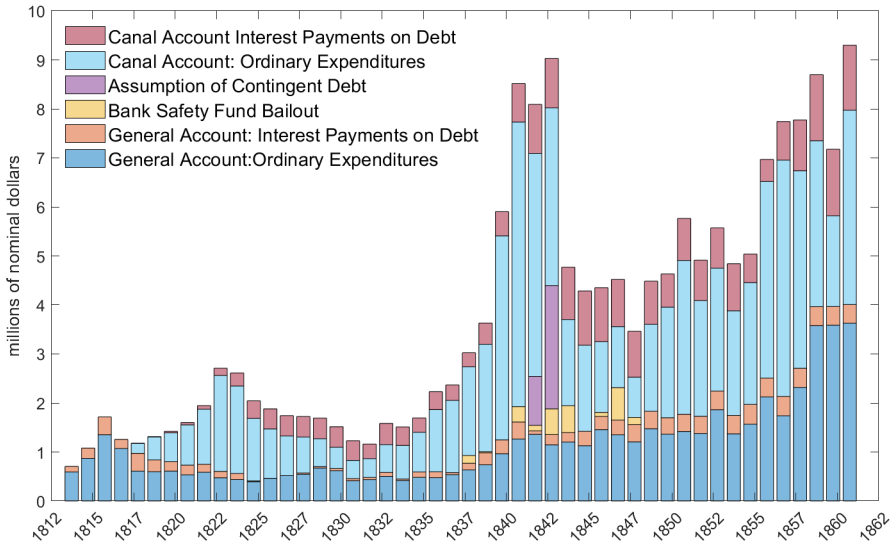

Figure 11: Consolidated Expenditures

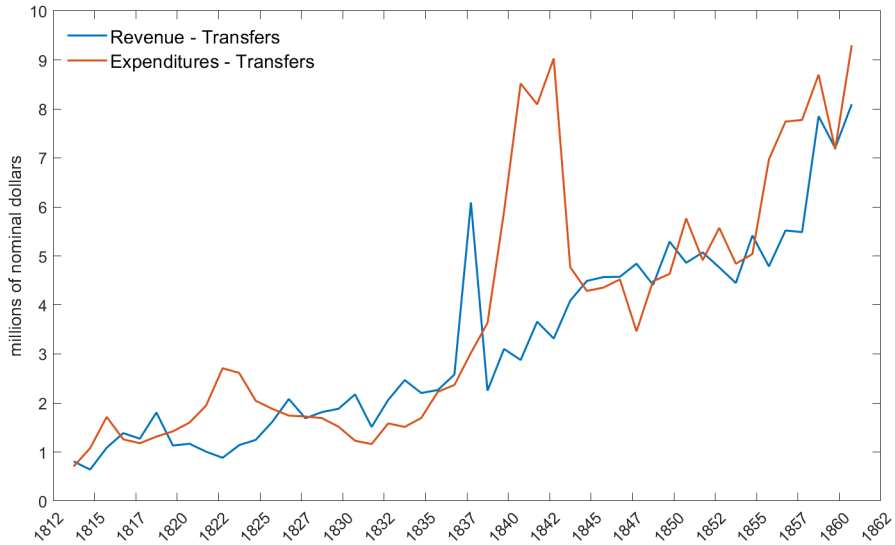

Figure 12: Consolidated Revenue and Expenditures

8 Bond Prices

- In figure 13 I plot the prices of three Canal bonds and three General Fund bonds.
- The late 1830s and the early 1840s were a period of fiscal stress for New York. During this period, the Legislature suspended all public works except those whose suspension would result in a significant loss and pledged the state's revenues for the payment of its obligations. It is hard to see this "suspension" in the Canal Account expenditures, figure 4.
- Despite never missing a coupon and principal payment, New York State bond traded at 80 cents on the dollar in the early 1840s. The Comptroller complains about this.
- I should convert prices to yields.

9 Unit of Account Issues

The source for this section is Roberts (1897).

- When specie payments were suspended during the financial panic of 1837, Comptroller Azariah C. Flagg used State resources to make up the difference between depreciated paper currency and coin to honor the State's obligations at par.
- In 1857, Comptroller William F. Allen likewise urged that the State's bonded debt be serviced in coin rather than in paper arguing that the public faith of New York should not

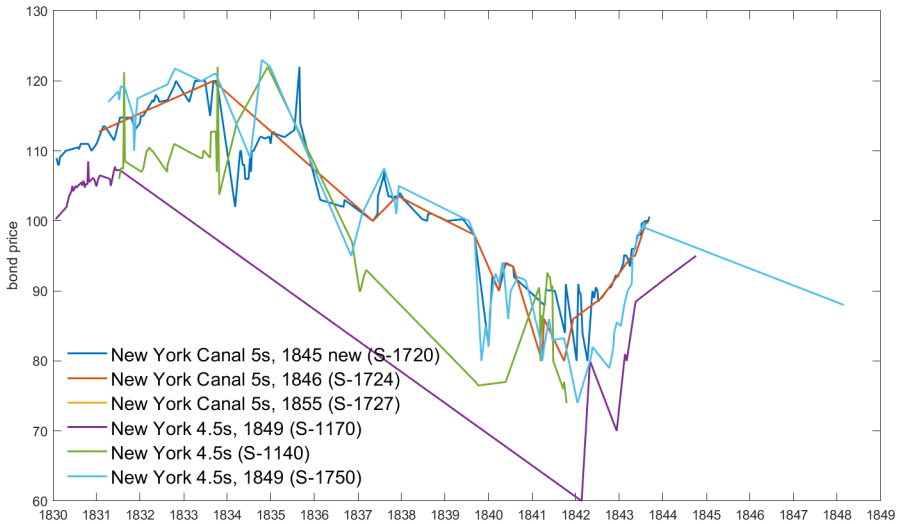

Figure 13: Prices of Various New York State Bonds

Source: Sylla et al. (2006).

fluctuate with the market value of its currency.

- During the Civil War, when specie commanded a substantial premium over paper currency, the State of New York faced the question of whether its own bonded obligations should be paid in coin or in depreciated legal-tender notes. In 1863 and 1864, Comptroller Lucius Robinson recommended that both the principal and the interest on the State's bonded debt be paid in specie. The The legislature declined to follow his advice.

- Although the Legislature again deferred action, the policy eventually prevailed. In 1870, the State entered the open market and purchased coin to pay interest on its outstanding bonds. This practice was continued until the general resumption of specie payments by the Federal Government in 1879.

- By adhering voluntarily to a metallic standard during the years of national paper currency, New York strengthened its standing in credit markets and set a precedent for conservative financial management that endured well into the late nineteenth century.

10 Data Sources

- The data for revenues and expenditures for the General Account are from Appendices I and II of Sowers (1914) and the *Annual Report of the Comptroller* for the State of New York.

The decomposition of General Fund revenues is from Sower.

- Data on the debt outstanding, fund balances, and Treasury balances are from the *Annual Report of the Comptroller*.
- The Canal Account revenues, expenditures, debt outstanding, and fund balances are from Table No 82 on page 88 of the 1860 *Annual Report of the Commissioners of the Canal Fund to the Legislature of the State of New York*.
- Unlike other state fiscal datasets, our records are consistent with the budget constraints presented at the start of this memo.

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